

CSC318H: The Design of Interactive Computational Media

A3: High-Fidelity Prototype and Usability Study -

Over-the-counter Medicine Finder:

MediSense

Studio 24

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High-Fidelity Prototype

In order to use our prototype, access the Figma prototype with this link for a [demo](#) and maximize the screen to enjoy the full experience. From here, you can click or use the mouse to hover over elements to interact with our prototype.

To access the full Figma prototype pages, you can do so by accessing the [view-only](#).

You will notice that this is an updated high-fidelity prototype. Updates have been made since the last feedback, so it may be a little different from the ones you see in the video.

Functional Requirements

From A1, we concluded the following 5 major functional requirements: equivalency explanation, price availability, mobile use, generic retrieval, and expert verification.

Equivalency Explanation

- For equivalency explanation, there are several ways for the user to determine whether the drug is equivalent or not.
- First, we provide one marketplace drug, a so-called “branded drug” in the top section of our recommendation. Following by, we provide the description “alternatives” and list out generic drugs which contain equivalent ingredients at a lower price. Users can click on these alternatives and find out the ingredients and compare them with the branded one.

- Importantly, we have added an “Equivalent Explanation” section, in which, user can expand more to read the explanation. It covers how it differs from the branded one and explains the equivalency of the chosen generic drug compared to the branded one.

Price Availability

- Offering an equivalent drug at a much lower price by a significant margin is where the core usefulness of our application lies. Thus, we achieve this by offering “generic drugs”, which contain similar or the same ingredients, effectiveness and use case, at a lower price without a premium price added.
- We list out the drugs in ascending order of price, and the user now can filter out the drugs by price range. This is a small yet powerful feature for users who are primarily looking for drugs within a price availability range.

Mobile Use

- Our prototype has been designed and focused primarily on mobile design. Mobile use isn’t just about the scale of design. It also needs to ensure the features and layout are very easy to use and interpret on mobile. We have optimized the layout as minimally as possible to avoid information jamming. We used common black and white themes throughout the design, maintained a concise, consistent layout, kept the flow of the use case simple, and buttons are intuitive to interpret and use.

Generic Retrieval

- First, we offer a variety of user input. This can be symptoms, the name of the brand, ingredients or a scan of a bar code of a brand or an image of a brand name on visible. Then, we list out our recommendations. One branded at the top for the sake of

comparison, and generic drugs under the name “alternatives”. Users can filter or sort based on price and other attributes.

Expert Verification

- Once a user selects a drug to look for more information, the user can click on an icon with an expert symbol. This leads to an expert verification page which contains an expert’s opinion on the drug. It will provide useful links for the user to research more and references that support the expert's claim.

How We Addressed the Issues Raised in A2

Several changes were made to address the issues raised in A2. Some of them were changed again after the in-person evaluation, receiving further feedback.

Enhanced Equivalency Explanation

- To deal with the lack of information when displaying the equivalency information, we made a couple of changes to the UI. First, to deal with the overall UI improvement and space jamming issue, we made changes to the search results. When users search, the results used to be divided into “marketplace” and “our recommendations” sections in A2’s prototype. This caused some confusion among the stakeholders. So, instead, we labeled this with “branded” and “generics”—later, “generics” was renamed to “alternatives” again. This rename made it clear for the user to distinguish between branded and generic drugs.
- From these new search results UI, when a user clicks on the drug, we now offer a tab for “Equivalency Explanation”. Back in A2, we made a proposal to add an

equivalency checkmark. Instead of jamming up the search results with a lack of space, we made another tab for equivalency explanation. In this tab, we provide a detailed explanation of how the generic drugs differ from or are similar to the branded drug.

- This covers the problem from A2, where we had a lack of explanation for medication equivalency. This effectively covers hesitation or doubt for the user when making a purchasing decision.

Streamlined Navigation, Filter, and Sorting

- Secondly, in the A2 prototype, there was an issue with navigation and search features. This usability issue was a combination of many problems. First of all, users had difficulty navigating different tabs. To improve this, in our prototype that we used for evaluation, we included the navigation bar. However, users gave feedback that the navigation bar indeed made things more confusing, and the UI was overwhelmed with information. Thus, to simplify the design, we have gotten rid of the navigation and pharmacist verification check mark by simplifying and reallocating it to different “tabs”.
- Users can now use the search bar to navigate to the home page, or when the user clicks on the drugs for more information, we provide four tabs, and among those tabs, users can switch back and forth to navigate those subpages. After the feedback, we have concluded that instead of adding a back button, this way of simplifying design in a more concise, consistent manner makes the navigation experience better.
- Adding on, more advanced search features were added to improve the search experience. In A2, we did not have a filter or sorting feature for search results. We proposed to add those as well as the search history feature. In our current design, we have added a filter option with “pharmacist verified only”, “name-brand only”, and

price range selection. For sorting, the user can use the cost option to sort in ascending order. We have added search history as well, promised in the A2 proposal. This allows users a more streamlined navigation and search experience.

Fixed Map Pinpoint, More Features, and Smooth Experience

- Third, in A2, there was concern that map features were lacking key functionality that is needed for the user to complete their end goal: purchasing the drug. We fixed the issue where the map does not allocate the user's location by adding a blue pinpoint. To improve the visual experience, we made the closest location of the place to buy the drug, as black, to make it distinguishable and highlighted. Other white buttons along with the price are used to represent other locations to purchase, but with a longer distance from the user's location.
- After our evaluation and receiving feedback, we made it easier for the user to jump to the address/direction page by making those black and white buttons clickable.
- Regarding the "address/direction" page, to make it easier for the user to complete their goal, which is to purchase a drug, we added two buttons in this page. One is "address", which allows users to copy the address, and another one is "direction". This allows the user to open the new Google Maps window.

Revised Product Information Layout

- Fourth, we have revised the product information layout. Instead of overwhelming the search result page with essential data—active ingredients, price, image—and with the supplementary details like usage instructions, warning, and pharmacist verification/explanation, we made those supplementary details accessible via an expandable section, integrated into tabs.

- When a user clicks on a drug from a search result, it brings them to pages with four tabs. From the first tab, the user can find out about the product, ingredients, usage, and equivalency explanation in an expandable and collapsible manner. In the second tab, the user can find information about the pharmacist verification along with the explanation and useful links. The third and fourth tabs are map-related, as mentioned in the second point.
- This improvement allows users to quickly scan for key information without being overwhelmed. Most importantly, after receiving feedback, reallocating the pharmacist verification checkmark from search results to those tabs was crucial to improve the information layout, as users had a hard time finding the page.

Detailed Expert Verification Explanation

- Lastly, connecting to the fourth point, we have improved the lack of information with expert verification. In our paper prototype, we merely had a pharmacist verification checkmark, and that was it. To support more detailed information, we have added a pharmacist's explanation page. This supports more information on generic drug safety including an expert's explanation, authoritative resources with links that back up the claim, and a green checkmark as a visual emphasis. This deals with the concern in A2, to build greater trust in generic drug recommendations within our application.

Evaluation Protocol

To assess the usability and effectiveness of our high-fidelity prototype, we designed a structured usability study that utilizes multiple data collection methods. This method was created to evaluate how well our prototype meets the goals, needs, and behaviors of our target users, and how effectively it can help with our original problem statement.

Participant Recruitment

We recruited 8 participants from the same studio as our group who closely match our target user base. All participants had prior experience purchasing over-the-counter (OTC) medications and had key characteristics of our intended audience, which include varying levels of familiarity with generic vs. brand-name options and different priorities about price and trust.

The participants were selected to include:

- Individuals who frequently purchase OTC medications and are comfortable comparing products
- Individuals who purchase infrequently and may rely more on brand recognition
- Users who prioritize price when purchasing or are curious about generic alternatives

Although the sample came from a single environment, we ensured diversity in medication purchasing habits and attitudes toward generic drugs. This allowed us to gather relevant, user-centered feedback during our evaluation while maintaining a controlled and consistent testing setup.

Furthermore, all usability sessions were conducted in-person, which allowed us to observe participants' real-time reactions, navigation patterns, and behaviors more directly. This setting also facilitated richer, more interactive feedback, as participants were more comfortable asking questions and providing commentary throughout the tasks. Conducting the study in person ultimately helped us collect more significant insights into both usability challenges and user expectations.

Tasks

Each participant was asked to complete four tasks that simulated real-world interactions with the app:

1. Search for a medication either using brand name, ingredient, or symptoms and view generic alternatives.
2. Use the product scanning feature to search for the medication.
3. Interact with the map and find locations to purchase the medication.
4. View pharmacist-verified information/explanation.

These tasks were chosen to evaluate the app's core functionalities and ability to support informed, cost-effective decisions, reflecting our original design requirements from A1.

Data Collection Methods

To gather comprehensive feedback to examine the usability of the design and the applicability of the design for the target users, we used four data collection methods across three data categories:

- Quantitative Objective
 - Task Completion Time: Measured how long each user took to complete each task
 - Error Rate: Counted the number of misclicks, backtracking, or help requests.
- Quantitative Subjective:
 - System Usability Scale (SUS): Collected at the end to assess overall usability using Google Forms.
- Qualitative Objective:
 - Navigation Observations: We recorded screen activity and noted where users struggled, hesitated, or seemed confused.
- Qualitative Subjective:
 - Think-Aloud Protocol: Participants were asked to verbalize their thoughts while completing tasks.
 - Post-Task Interviews: Short, open-ended questions were used to gather user reflections on each task and the overall experience.

Procedure

Before we began the evaluation, we made it clear to the participant that we may use the data from the study and may take a video, which may be used for the report. The Google Forms were used to collect name, email, and consent for participating in Studio 24's usability study.

Then, we described the task to conduct and began the study and recorded the data according to the collection methods.

Rationale for Study Protocol and Data Collection

- The rationale behind our study protocol was to simulate realistic scenarios where users would interact with the app when searching for affordable and trustworthy OTC medications. By designing tasks that align closely with our core functional requirements, we were able to assess how well our prototype supports informed, efficient, and cost-effective decision-making. We chose a mix of quantitative and qualitative methods to capture both measurable usability metrics and deeper insights into user thought processes.
- We measured task completion time to ensure our product helps with “quickly finding accurate information,” which was part of our problem statement. Our product would not be desirable if the user cannot complete tasks promptly. Error rate was also essential to include in the study protocol in order to measure how straightforward our UI is to navigate.
- Using the System Usability Scale (SUS) gave us a standardized measure to compare usability across participants. Furthermore, the inclusion of think-aloud protocols and post-task interviews helped us understand users’ thought processes and expectations, while screen recordings allowed us to detect usability issues that are not always revealed through self-reported data.

Overall, this mixed-method approach allowed us to evaluate not only how users interacted with the system but also how confident and satisfied they felt throughout the experience,

directly tying back to our goals: to make it easy to quickly find accurate information about affordable OTC medicine.

Study Results

*Some quotes are repeated in the evidence. In those cases, we won't repeat the quote, instead, just reference the video number and timestamp. You can find all our clips and videos for reference in Google Drive.

Usability

1. Users found several UI elements in the prototype to be misleading.

Study participants found several UI elements in our high-fidelity prototype to be misleading or ambiguous in meaning. The colours and symbols chosen were confusing to the usability study participants. Specifically, the following UI elements were mentioned the most by the participants:

- An indicator for whether a medication is name-brand or an alternative brand, and
- Button for the page showing a list of nearby pharmacies.

Evidence

- One participant asked, “So this is branded... is that not good, or...?” upon first seeing the symbol chosen to represent name-brand products in the search results, a red octagon with an exclamation mark ( a3clip1.mp4). Another participant also indicated confusion, stating that it looks more “like a warning” rather than regular information (Video 5, 00:34).
- For the button to access the page of nearby pharmacies, multiple participants said that they associate the shopping cart symbol more with checking out at an online retailer. For instance, “shopping cart as an icon doesn't make too much

intuitive .. because I associate it with online shopping” (Video 8, 2:50). This confusion led to significant navigation difficulties in Task 3, finding purchase locations, with users struggling to find the directions page, and suggesting a “Pin point icon...instead of [checkout] icon” (Video 3, 2:50).

- Some users expected map pinpoints themselves to be clickable for directions, “I was expecting if I press onto this thing [Price icon on map] it would show this page [pharmacy address & directions page that appears when pressing shopping cart icon]” (Video 7, 2:43). All participants made at least one error while navigating to this page, with one participant making 6 extra clicks before finding this page.

Importance

Nielsen’s usability heuristic number 6 states that recognition is better than recall.

Having ambiguous symbology in the app hurts the usability of our app since it forces the user to remember what each symbol means, rather than recognizing its meaning from the graphic itself. This adds an additional barrier in the usability because infrequent or first-time users would not be able to use the app efficiently without external help. In addition, having misleading UI elements increases the possibility that the information presented in our app is not communicated correctly to the end user.

Since our app deals with products that have serious health consequences when misused, we cannot afford miscommunication.

2. Users found it difficult to distinguish between clickable and non-clickable UI elements

Study participants had a hard time determining whether a UI element was clickable as a button, specifically during Task 4, finding pharmacist-verified information. While it

changes colour when hovered over, the green check mark next to the brand name did not appear obviously clickable to participants.

Evidence

- Task 4 had the highest number of mistakes made across all participants, with an average 4 mistakes made per participant during the performance of this task (See Figure 1). Compared to Tasks 1 and 2, this is an increase of more than 3 mistakes.
- Two participants could not get to the correct page without intervention from one of our group members ( a3clip2.mp4). Multiple participants indicated that they did not think to check if the green tick mark is clickable, “I did not realize they had to click green checkmark to get pharmacist verified information” (Video 7, 4:10). One participant said they “don’t see [anything] that highlights whether it’s pharmacist verified” when asked to perform this task (Video 8, 4:00).
- Suggestions included adding text next to the icon or using underlining, “...not super clear why this [the icons Branded and Generics] is not clickable but this [checkmark icon] is clickable...Maybe text underline”, (Video 3, 04:21).
- Another suggested needing a legend or explicitly labelling the icon, “I think maybe bring it to this side [below the Branded icon] might be a good idea as they are both clickable” and “Thumbs up is a bit confusing ... I guess if it just had the generic with no thumbs up I’d be more likely to click into this ... easier to identify if it says pharmacist verified next to the checkmark” (Video 4, 06:01; Video 7, 5:22).

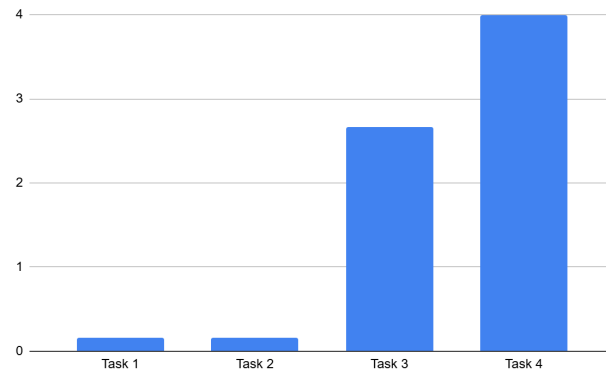


Figure 1. Average number of errors made by a participant per task.

Importance

Expert verification is one of our product's five design requirements. Therefore, it is important for the users of our product to be able to easily access the page on pharmacist verified information. This information is the defining feature of our product, since pharmacists are a voice of authority when it comes to OTC medications. Having a clearly defined, clickable button to bring the user to the page containing this information improves the usability of our app by streamlining the process and reducing both the time it takes for a user to find the page and the chance for potential user errors.

Applicability

3. Users were able to efficiently find information about relevant medication

Study participants generally had no problems using the prototype to look for relevant medication, either through the search bar function or the camera scan function. Across every participant, the search procedure was efficient and minimized user error.

Evidence

- As seen in the previous figure of the average number of mistakes made, the two tasks involving searching for relevant medications achieved a significantly lower number of errors, which means that all participants were able to learn to perform this task on our prototype without prior knowledge.
- In addition, the average completion time of a search using the search bar was 42.51 seconds, while the average completion time of a search using the camera was 13.34 seconds. Multiple participants indicated that the search function was “easy to navigate”. Minor suggestions included highlighting the scanned item more clearly, “maybe highlight this is specifically what you have scanned” (Video 8, 1:30).
- Lastly, all except one study participant gave our prototype a System Usability Score of over 70 as shown in Figure 2, with the average being 72.5. Taking all of this into consideration, we can confidently say that users can quickly and efficiently use our prototype to search for alternatives to name-brand drugs and find information relating to the products.

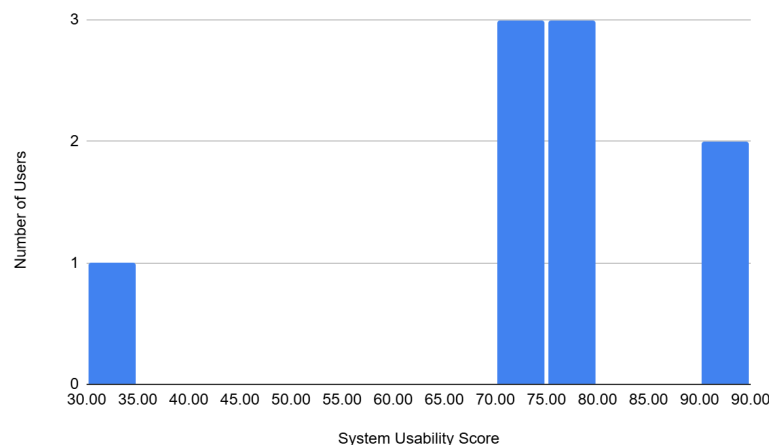


Figure 2. Histogram of System Usability Scores given to the prototype by study participants.

Importance

Efficiently finding accurate information about OTC medication products is the problem we are trying to tackle with this product. Making the search process as straightforward and painless as possible is important for multiple reasons. First, it reduces the barrier of entry to using our product. Since the first action that users will do in our app is search for medication, a good first impression encourages them to further explore other functionalities of the app, making full use of our product. Secondly, it helps solve the problem outlined by our problem statement. By making the searches quick, efficient, and accurate, the consumer will be empowered to make choices about their purchases based on information that was previously difficult to access in a timely manner or obscured by corporate advertising.

4. Lack of Clarity on Core Terminology, “Generic” Limits Understanding

During the study, some of our users expressed uncertainty or a lack of understanding about our app’s fundamental concept of “generic” medications. While most of our participants were already familiar with our application and concepts, through several presentations and studies beforehand, this indicates a potential gap in the app’s ability to effectively educate and empower all users.

Evidence

- During our study, at least one participant has explicitly asked, “what do you mean by generic?” ( a3clip3.mp4). While the interface uses labels like “Branded” and “Generics”—now we renamed “Generics” to “Alternatives”—the direct question suggests the underlying concept might not be universally understood by the target audience.

- In addition, the misinterpretation of the red octagon symbol for “branded” products, for instance, a participant asking “So this is branded... is that not good, or...?” suggests the label “Branded” alone paired with this symbol was not sufficient to clearly convey the meaning of the expensive premium price that comes with branded drugs ( a3clip1.mp4).
- This also implies that the distinction between categories like “Branded” and “Generics”—now renamed to “Alternatives” wasn’t obvious or intuitive. This lack of clarity on core terminology may potentially hinder users’ confidence in choosing alternatives.

Importance

For the app to be truly applicable in helping users make informed, cost-effective choices, as we stated in A1, users must be able to understand the distinction and equivalency between branded and generic options. If the core terminology, “generics,” is unclear to the users, the app may fail to fully address the information gap we intended to solve. This highlights the need for defining terms within the app clearly with accessible, simple terms or more detailed explanations within our app.

5. Prototype Does Not Fully Bridge the Trust Gap Needed for Users to Confidently Choose Generic Alternatives

While the prototype includes features designed to build trust, like the features we added: expert verification and equivalency explanations, the overall presentation and accessibility of this information, combined with user uncertainty about the core concepts, implies that the prototype is currently failing to effectively convince users and bridging the confidence gap regarding the generic medications. One of our major

applications of our app was to tackle the problem of people's lack of knowledge and trust in generic drugs, and by doing so, make purchasing drugs easy, quick, and affordable. But it appears the features we have are not sufficient to fully convince or educate the users. It's not that our features are ineffective, just that it's not enough to fully bridge the trust gap needed for users to confidently choose generic alternatives.

Evidence

- The core problem the app aims to address is the user's hesitation towards generic drugs due to perceived risks or lack of trusted information, as supported by findings from A1. But the in-person evaluation shows that key features meant to address this are not effectively utilized or understood.
- For instance, though this may link more towards the usability, the fact that users struggled significantly to even find the Pharmacist Verification information, "I realized, I don't really see a thing that highlights whether it's pharmacist verified...?" (Video 8, 4:00), and more examples we showed already in Video 3, 04:21 and Video 7, 4:10, indicates this trust signal wasn't successfully communicated.
- Furthermore, fundamental confusion persisted for some users regarding the meaning of "generic" ( a3clip3.mp4) and the interpretation of symbols differentiating branded and generics ( a3clip1.mp4 ; Video 7, 5:22).
- Adding to this challenge in building trust, the study revealed deeper confusion about the very meaning and value of the 'pharmacist verified' designation itself, with one participant stating, "Overall like... confused .. like all of them should be like pharmacists verified anyways... I don't know exactly what

pharmacist-verified means” (Video 5, 9:50), indicating the feature failed to convey its specific significance within the app’s context.

- Suggestions for more explicit labels like “pharmacist verified” next to the checkmark (Video 7, 5:22) highlight that users require clearer, more direct assurances than currently provided to feel confident. Even with some of our features available, designed to deal with narrowing the gap in trust, our high-fidelity prototype hindered in their effectiveness in assuring users.

Importance

As mentioned, our application’s applicability relies on its ability to empower users to make informed, cost-effective choices, which requires overcoming and dealing with skepticism about generic drugs. If users finish interacting with the prototype still feeling uncertain about the safety, equivalency, or trustworthiness of alternatives due to confusing symbols, unclear terminology, or inaccessible verification data, then it indicates the high-fidelity prototype has not fully solved the target problem. This has been shown in our study results, failing to fully solve the problem we intended to solve. The application needs further improvement to support the user’s decision-making process by effectively narrowing the trust gap needed for users to confidently choose generic alternatives.

Lessons Learned and Implications for Design

Our usability study provided valuable insights on how users interact with our high-fidelity prototype and revealed challenges in both interface design and communication of core concepts. Critical issues that limit the app's ability to educate, build trust, and enable informed decision-making about generic medication were identified. These lessons will guide the refinements we implement for our final demo in A4.

Lesson 1: Symbols and Icons Must Be Instantly Recognizable

Several UI elements were continuously misinterpreted, particularly the red octagon for branded products and the shopping cart icon for finding nearby pharmacies. Users found them misleading or confusing, and they often associated them with warnings or online checkout rather than neutral or helpful information.

We will redesign these icons using standard, widely recognized symbols. For instance:

- Replace the red octagon with a neutral label or pill icon to represent branded medications.
- Replace shopping cart icon with map pin to better reflect its indented function.

Lesson 2: Clickable Elements Need Clear Indications

Participants consistently struggled to distinguish between clickable and non-clickable UI elements, especially the pharmacist verification checkmark. Although this feature is an important trust-building component, users failed to realize it was interactive.

We will improve the affordance of interactive elements by:

- Adding text labels (ex. "Pharmacist Verified") next to icons
- Using hover animations or button styling to visually indicate clickability

Lesson 3: Concepts Like “Generic” Must Be Clearly Defined

Although some participants were already familiar with the concept of generics due to earlier exposure, several participants still expressed confusion or asked for clarification during the study. This indicates a potential knowledge barrier for new users unfamiliar with the concept of medication equivalency.

We will introduce:

- “What is a Generic?” info button at the top of relevant pages
- A short tour of UI elements to improve first-time user understanding
- Add short onboarding screen explaining branding vs. generic medications and equivalency.

Lesson 4: Trust-Building Features Need More Emphasis and Clarity

Participants appreciated the idea of expert verification but found its current placement and label too unclear. Some did not understand what “pharmacist verified” meant or why it mattered.

We will emphasize the labels by:

- Moving the “Pharmacist Verified” icon to a more visible position
- Adding clearer language to explain what verification means and why it is important

Lesson 5: Navigation Should Be More Intuitive and Guided

Users made multiple navigation errors when trying to find specific information, especially directions to nearby pharmacies. This indicates a need for more visual guidance.

We will:

- Make map pins directly clickable

- Highlight relevant features like address copy and Google Maps links with clear icons and labels.

Here is a summary of our planned design changes:

Problem Identified	Planned Change
Misleading red octagon symbol	Replace with pill icon/neutral icon
Confusing shopping cart icon	Replace with map pin
Poor visibility of clickable icons	Add labels, hover effects, etc
Confusion about “generic” terminology	Add info page
Weak emphasis on verification	Move and relabel checkmark with explanation
Navigation issues with map	Make pins clickable

Reflection

What Went as Expected

From the start, we expected that price transparency would be the primary value driver for our app, and that simplifying the process of comparing medications would immediately resonate with users. During the early research phase and low-fidelity prototyping, this assumption seemed to hold true. Many participants confirmed they found it frustrating to navigate between brand-name and generic products without centralized information. As a result, when we incorporated features like price sorting, product equivalency, and the camera scan function, participants found them helpful and easy to use.

We also anticipated that studio-based iteration would help us quickly improve our design. This proved true. Getting continuous feedback from peers and instructors allowed us to identify and address UI issues in every milestone. Our navigation bar, for example, was tested and iterated multiple times before being replaced with the current tab structure. In that sense, the rapid prototyping cycle matched our expectations, and we were able to produce a high-fidelity prototype with multiple refined flows.

What Did Not Go as Expected

What we didn't anticipate was how central the issue of trust would become. While we focused early on affordability and efficiency, many participants during field studies and evaluations expressed a much stronger concern for safety, often defaulting to brand-name medications simply because they were familiar. Several even stated they were willing to pay more if it meant they could trust the product. This insight was a major turning point: it shifted

our framing of the problem from “price comparison” to “bridging the trust gap between branded and generic medications,” though an affordable drug was still within our radar.

We also underestimated how challenging it would be to convey trust and safety through UI alone. Symbols we thought were clear, like a green checkmark for verification or a red octagon for brand-name drugs, ended up being misinterpreted. The shopping cart icon, which we originally used to link users to pharmacies, was associated with online checkout rather than physical locations. These misunderstandings revealed how easily well-intentioned UI choices can backfire when the mental model doesn’t align with the user’s expectations.

Another unexpected challenge was how much terminology mattered. Words like “generic” weren’t always understood, and even the label “pharmacist verified” raised questions about what that really meant. We had assumed that most users would have a baseline familiarity with these terms, but usability testing showed us otherwise.

What We Would Do Differently

If we were to start the user-centered design process over again, one of the biggest changes we’d make is prioritizing concept validation earlier in the process. Before committing to a full UI design, we would test how users interpret our core terminology (“generic”, “verified”, etc.) and key symbols. Doing this upfront would have helped us avoid several late-stage miscommunications and usability issues.

We also would have spent more time early on exploring the emotional landscape around medication decisions. What motivates users to choose one product over another? What emotional or psychological reassurance are they seeking? These questions would have

pushed us to design trust-building features, like pharmacist bios, user testimonials, or ratings, sooner, rather than introducing them reactively after usability issues surfaced.

Additionally, we would build in onboarding and educational components from the very first iteration. Even simple introductory tooltips or a short “Why Generics?” explainer could have helped users better understand our product’s value and reduced confusion during testing.

These types of features weren’t prioritized in our initial roadmap, but we now see how essential they are for clarity, especially for first-time users.

Finally, we would have conducted more low-fidelity A/B testing for alternative UI components (e.g., different verification icons, layout flows) to de-risk usability issues before committing to visual design polish. A few of our late-stage pivots, such as replacing icons or adjusting page flows, could have been caught and validated earlier with simpler wireframe tests.

Appendices

[1] High-Fidelity Prototype Demo

<https://www.figma.com/proto/J3FbKFLrcmAdf4kx1frDUP/CSC318?node-id=2003-274&p=f&t=AikijfGwaxH5Bxo-1&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=2003%3A274>

[2] Figma Pages

<https://www.figma.com/design/J3FbKFLrcmAdf4kx1frDUP/CSC318?node-id=2003-65&m=dev&t=5Z9sDgl6pdjyKF2b-1>

[3] Usability Study Videos and Clips

https://drive.google.com/drive/folders/1AD7Ai0cW500_MwTFVelhKcCHSSlx2vKn?usp=sharing

[4] Usability Study Raw Data

https://docs.google.com/spreadsheets/d/15mAhhB1LeUcCac8Z7tVMq-r8xUAcoLgcR7lh7qkF7T_c/edit?usp=sharing

[5] Usability Consent Form

<https://drive.google.com/file/d/1eLeocPoptMZBJR9dan5KUxXaxt2YIb9/view?usp=sharing>

[6] Group Meeting Note by Yehyun

<https://drive.google.com/file/d/1YiIP2DADASyjf5kWBnd-Jh-pyb1CFiE1/view?usp=sharing>